

Fuyu Precision Component Co., Ltd

Testing Center

Dye & Pry Test Report

Report No: VN20260729007
Sample Name: PCBA
Applicant: ANBU II ME
Test Environment: 24.3°C/54.4%RH
Date of Received: Jul. 31, 2026
Date of Start: Aug. 1, 2026
Date of Completion: Aug. 3, 2026
Date of Report: Aug. 4, 2026

Approved by	Checked by	Prepared by
	 Ngô Trí Dũng	 Lê Minh Dũng

Notes:

- 1.This report is only applicable to the submitted samples.
- 2.Partial copy this report is prohibitive if not approved by testing center.

Add: Floor 1F & 2F, Building F01 of Lot M1, Lot F and Lot T1, Quang Chau Industrial Zone, Nenh Ward, Bac Ninh Province, Viet Nam

Tel: +84-222-3856 888 Ext: 33499/33495

1.Problem Description:

To verify the soldering quality, dye & pry test was performed to check for potential defects in the solder joints. The specimen to be tested was specified as below.

2.Sample Information:

No.	Customer	Sample Name	P/N	S/N	Location
#1	NVIDIA	PCBA	699-9X9OE-0300RC1N	2615V757977403	U1

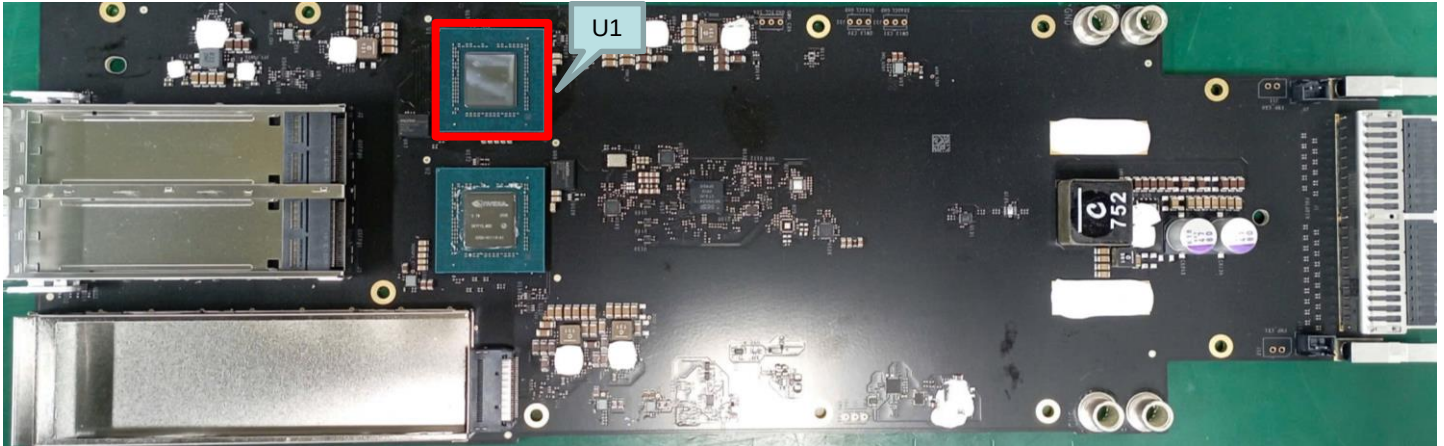
3.Test Equipment:

No.	Equipment Name	Brand	Model	Next Cal. Date
1	Microscope	Keyence	VHX-S550E	N/A
2	Vacuum system	Buehler	20-1382-250	N/A
3	Oven	Dr. Storage	AH-72	Jun. 21, 2027
4	Universal Testing Machine	KQL	KD111- 5	N/A

4.Test Procedure:

The dye & Pry test preparing procedure is conformed to "Dye & Pry Test Operation Process(SQ-3NVA-082)" and IPC-TM-650/2017 2.4.53.

5. Test Location:



6. Test Result:

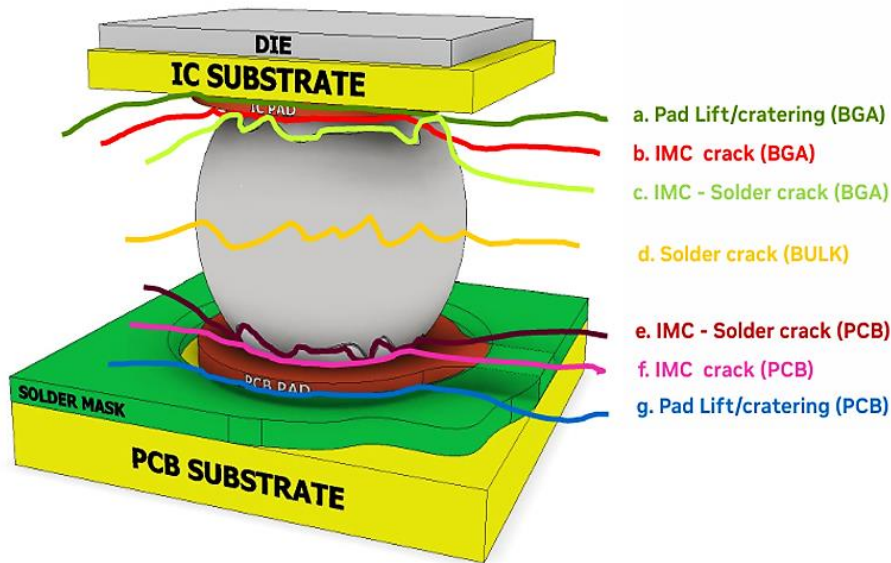
Sample No.	Location	Test Result	Remark
#1	U1	No dye penetration	Page 5~6

7. Dye and Pry Failure Mode Classification:

7.1 Failure Mode and Cracking Score:

Failure Mode:

Cracking Score:



0	= no dye penetration (crack area 0)
1	= dye penetration/crack area 1%-15%
2	= dye penetration/crack area 16%-25%
3	= dye penetration/crack area 26%-35%
4	= dye penetration/crack area 36%-45%
5	= dye penetration/crack area 46%-55%
6	= dye penetration/crack area 56%-65%
7	= dye penetration/crack area 66%-75%
8	= dye penetration/crack area 76%-85%
9	= dye penetration/crack area 86%-95%
10	= dye penetration/crack area 96%-100%

7.2 Pass/Fail Criteria:

Table 2 Acceptance Criteria of DnP

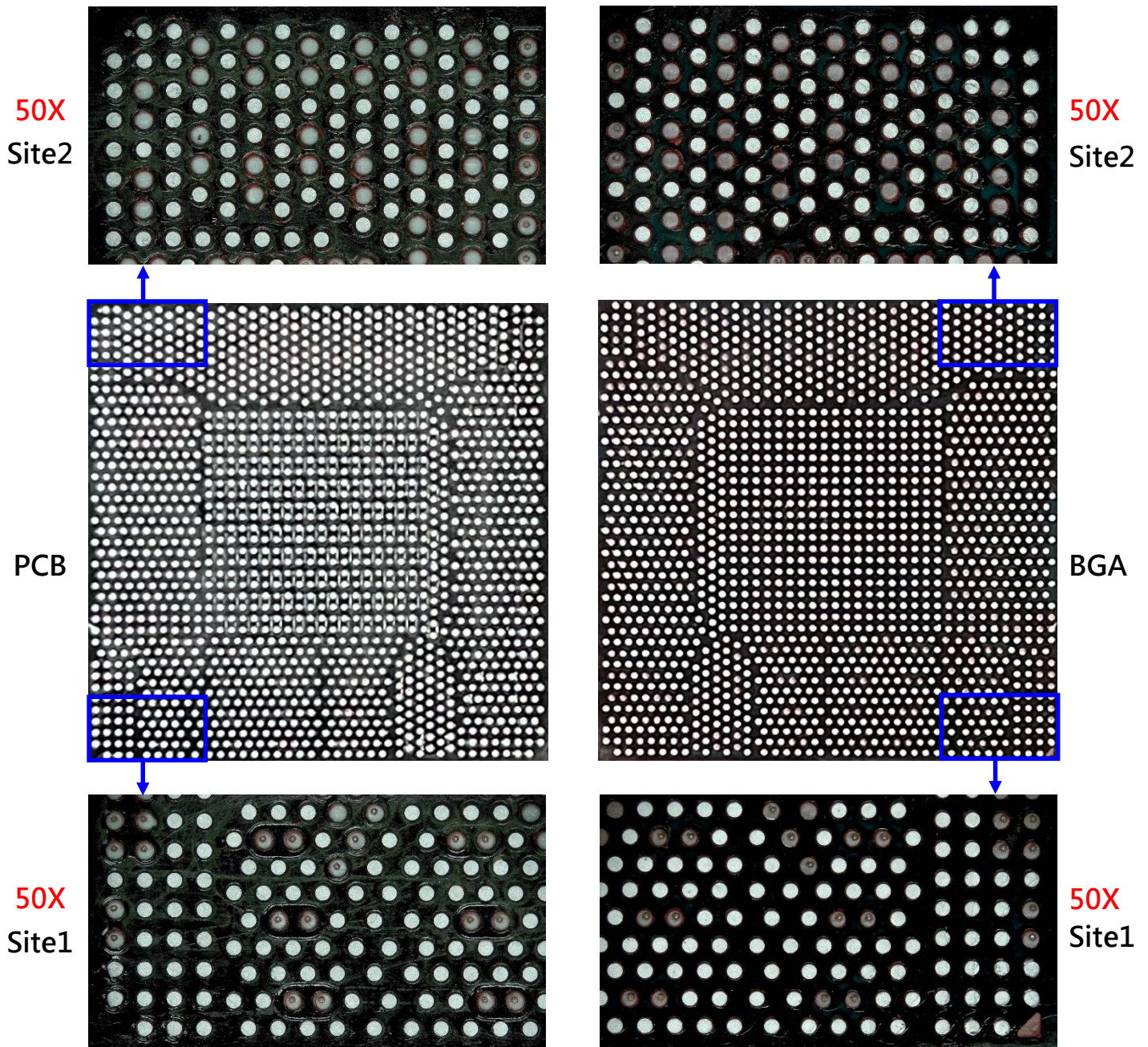
Table 2A NV GPU Card, Server and other Non-Auto PCBAs--DnP Criteria (FCT Pass)												
Pin	Process	Pass			Flag				Fail			
		BGA Solder Joint Cracking	PCB Pad Cratering		BGA Solder Joint Cracking	PCB Pad Cratering			BGA Solder Joint Cracking	PCB Pad Cratering		
			Non-VIPPO	VIPPO		Non-VIPPO	NC Pin	VIPPO		Non-VIPPO	NC Pin	VIPPO
Corner GND	Post Assembly	0,1	0,1,2	No via Crack	2	3,4,5	5,6,7,8,9	NA	≥3	≥6	10	Via Crack
	Post Non-shippable ORT Or Long-term stress	≤9	≤9	No via Crack	10	10	10	NA	NA	NA	NA	Via Crack
Signal & Other GND Pin	Post Assembly	0	0,1	No via Crack	1	2	5,6,7,8	NA	≥2	≥3	9,10	Via Crack
	Post Non-shippable ORT Or Long-term stress	≤6	≤6	No via Crack	6 < ~ ≤9	6 < ~ ≤9	NA	NA	10	10	NA	Via Crack
Action		No Action			Need Highlight and Bug report to track							

Table 2B NV Automotive PCBAs--DnP Criteria (FCT Pass)												
Pin	Process	Pass			Flag				Fail			
		BGA Solder Joint Cracking	PCB Pad Cratering		BGA Solder Joint Cracking	PCB Pad Cratering			BGA Solder Joint Cracking	PCB Pad Cratering		
			Non-VIPPO	VIPPO		Non-VIPPO	NC Pin	VIPPO		Non-VIPPO	NC Pin	VIPPO
Corner GND	Post Assembly	≤5%	≤10%	No via Crack	5% < ~ ≤15%	5% < ~ ≤25%	≥50%	NA	>15%	>25%	≥50%	Via Crack
	Post Non-shippable ORT Or Long-term stress	≤50%	≤50%	No via Crack	50% < ~ ≤85%	50% < ~ ≤85%	NA	NA	>85%	>85%	NA	Via Crack
Signal & Other GND Pin	Post Assembly	≤5%	≤10%	No via Crack	5% < ~ ≤15%	10% < ~ ≤15%	15% < ~ < 50%	NA	>15%	>15%	≥50%	Via Crack
	Post Non-shippable ORT Or Long-term stress	≤25%	≤50%	No via Crack	25% < ~ ≤50%	50% < ~ < 85%	50% < ~ ≤90%	NA	>50%	>85%	>90%	Via Crack
Action		No Action			Need Highlight and Bug report to track							

8. Dye & Pry Microstructure Inspection:

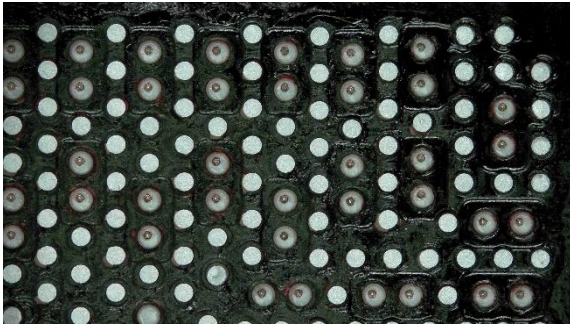
U1: No dye penetration. The representative photos are shown as below.

Site1 and Site2

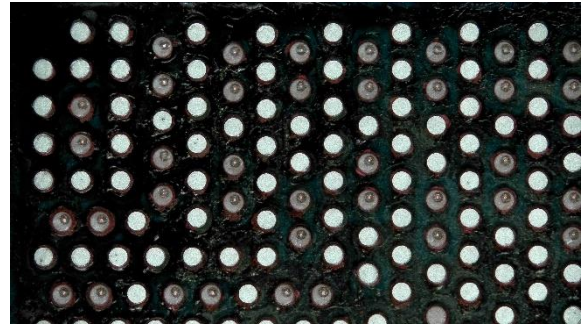


Site3 and Site4

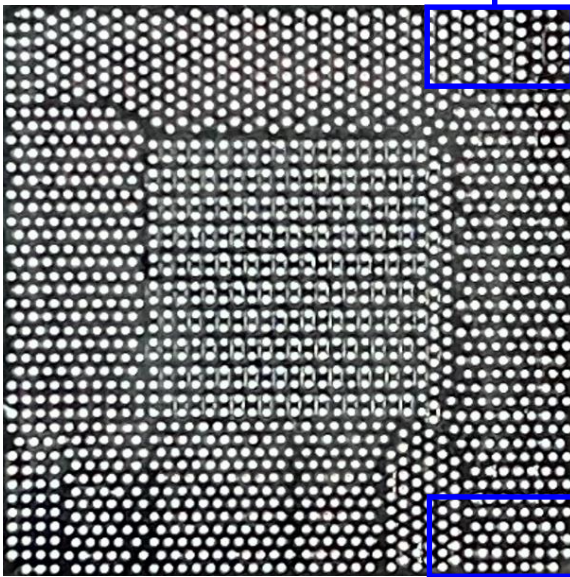
50X
Site3



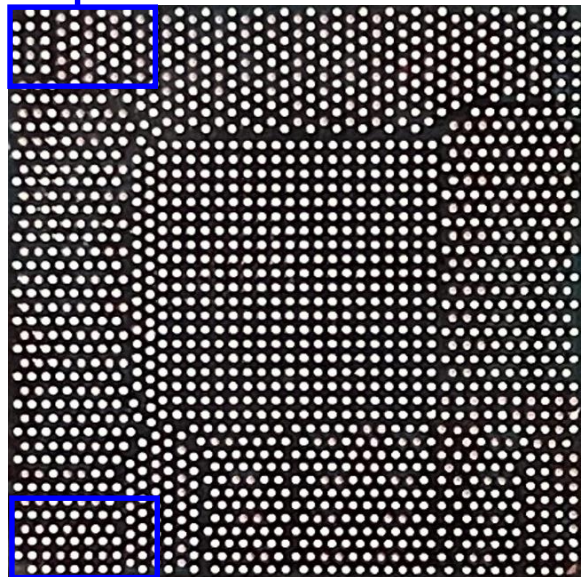
50X
Site3



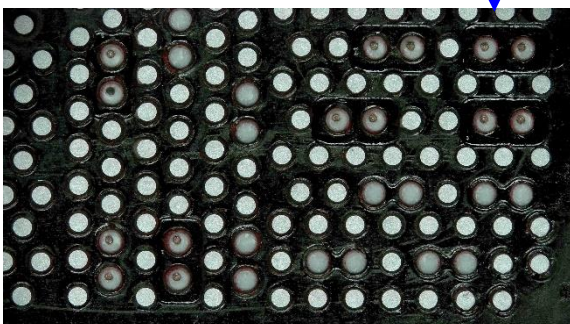
PCB



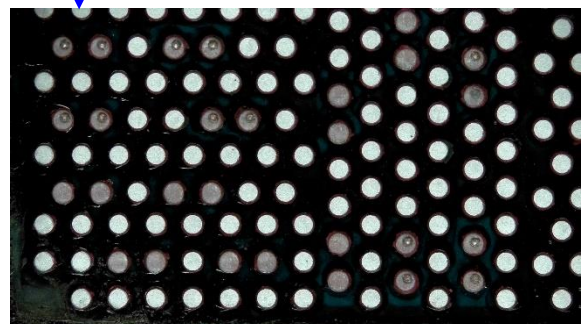
BGA



50X
Site4

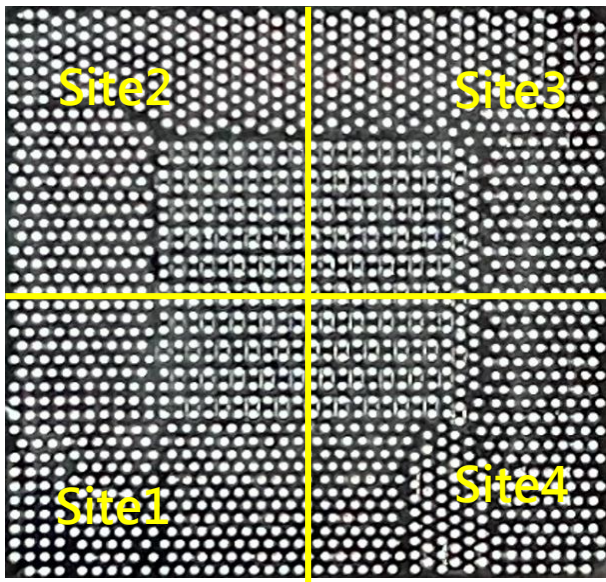


50X
Site4



9. Description of The Photo Sites:

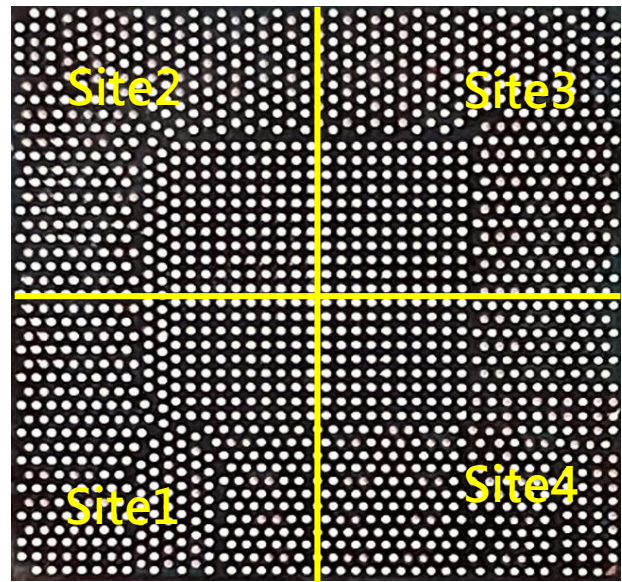
U1



PCB Side



Mark

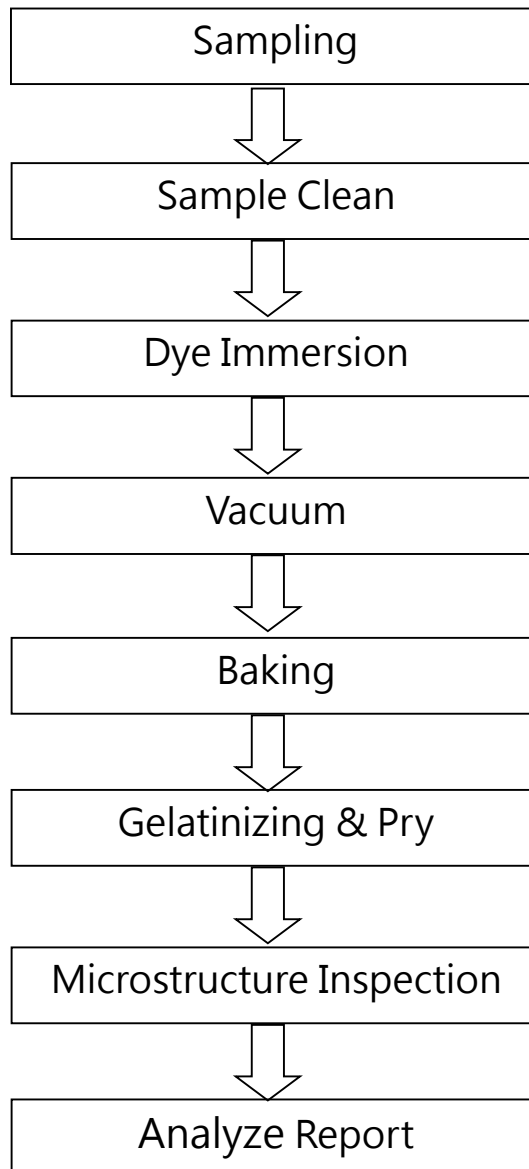


BGA Side



Mark

10. Dye & Pry Test Flow Chart



-----The End-----